

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only

Efavir-600 mg Tablet

Efavirenz Tablets USP 600 mg

COMPOSITION

Each film coated tablet contains:

Efavirenz USP 600 mg

Colour: Yellow Oxide of iron, Titanium Dioxide

DOSAGE FORM

Oral, film-coated tablet

DESCRIPTION

Yellow Colored, Capsule shaped biconvex tablets debossed with “EFV” on one side and plain on other side.

PHARMACOLOGY

Pharmacodynamics

Efavirenz (EFV) is a non-nucleoside reverse transcriptase inhibitor (NNRTI) of human immunodeficiency virus type 1 (HIV-1). Efavirenz activity is mediated predominantly by non-competitive inhibition of HIV-1 reverse transcriptase (RT). HIV-2 RT and human cellular DNA polymerases alpha, beta, gamma and delta are not inhibited by efavirenz.

Cardiac Electrophysiology

The effect of Efavir on the QTc interval was evaluated in an open-label, positive and placebo controlled, fixed single sequence 3-period, 3-treatment crossover QT study in 58 healthy subjects enriched for CYP2B6 polymorphisms. The mean C_{max} of efavirenz in subjects with CYP2B6 *6/*6 genotype following the administration of 600mg daily dose for 14 days was 2.25-fold the mean C_{max} observed in subjects with CYP2B6 *1/*1 genotype. A positive relationship between efavirenz concentration and QTc prolongation was observed. Based on the concentration-QTc relationship, the mean QTc prolongation and its upper bound 90% confidence interval are 8.7 ms and 11.3 ms in subjects with CYP2B6*6/*6 genotype following the administration of 600 mg daily dose for 14 days. (see **Section WARNINGS AND PRECAUTIONS & Section DRUG INTERACTIONS**).

Pharmacokinetics

Absorption

Peak efavirenz plasma concentrations of 1.6–9.1 μM were attained by 5 hours following single oral doses of 100 mg to 1600 mg administered to uninfected volunteers. Dose-related increase in C_{max} and AUC were less than proportional, suggesting diminished absorption at higher doses.

In HIV-1-infected patients at steady state, mean C_{max} , mean C_{min} , and mean AUC were dose proportional following 200 mg, 400 mg, and 600 mg daily doses. Time-to-peak plasma concentrations were approximately 3–5 hours and steady state plasma concentrations were reached in 6–10 days. In 35 patients receiving efavirenz 600 mg once daily, steady-state C_{max} was $12.9 \pm 3.7 \mu\text{M}$ (mean $\pm$ SD), steady-state C_{min} was $5.6 \pm 3.2 \mu\text{M}$, and the AUC was $184 \pm 73 \mu\text{M}\cdot\text{h}$.

Effect of Food on Oral Absorption

Tablets: Administration of a single 600 mg efavirenz tablet with a high-fat /high caloric meal (approximately 1000 kcal, 500–600 kcal from fat) was associated with a 28% increase in mean AUC(infinity) of efavirenz and a 79% increase in mean C_{max} of efavirenz relative to the exposures achieved under fasted conditions (see ***DOSAGE AND ADMINISTRATION***).

Distribution

Efavirenz is highly bound (approximately 99.5–99.75%) to human plasma proteins, predominantly albumin. In HIV-1 infected patients (n=9) who received efavirenz 200–600 mg once daily for at least 1 month, cerebrospinal fluid concentrations ranged from 0.26% to 1.19% (mean: 0.69%) of the corresponding plasma concentration. This proportion is approximately 3-fold higher than the non-protein-bound (free) fraction of efavirenz in plasma.

Metabolism

Studies in humans and in vitro studies using human liver microsomes have demonstrated that efavirenz is principally metabolized by the cytochrome CYP450 system to hydroxylated metabolites, with subsequent glucuronidation of these hydroxylated metabolites. These metabolites are essentially inactive against HIV-1. The in vitro studies suggest that CYP3A4 and CYP2B6 are the major isozymes responsible for efavirenz metabolism. Efavirenz has been shown to induce CYP450 enzymes, resulting in the induction of its own metabolism. Multiple doses of 200–400 mg per day for 10 days resulted in a lower than predicted extent of accumulation (22–42% lower) and a shorter terminal half-life of 40–55 hours (single dose half-life: 52–76 hours).

Elimination

Efavirenz has a terminal half-life of 52–76 hours after single doses and 40–55 hours after multiple doses. A 1-month mass balance/excretion study was conducted using 400 mg per day with a ¹⁴C-labeled dose administered on Day 8.

Approximately 14–34% of the radiolabel was recovered in the urine and 16–61% was recovered in the feces. Nearly all of the urinary excretion of the radiolabeled drug was in the form of metabolites. Efavirenz accounted for the majority of the total radioactivity measured in the feces.

Special Populations

Hepatic Impairment: The pharmacokinetics of efavirenz has not been adequately studied in patients with hepatic impairment (see **WARNINGS AND PRECAUTIONS**).

Renal Impairment: The pharmacokinetics of efavirenz has not been studied in patients with renal impairment; however, as less than 1% of efavirenz is excreted unchanged in the urine, the impact of renal impairment on efavirenz elimination should be minimal.

Gender and Race: The pharmacokinetics of efavirenz in patients appears to be similar between men and women and among the racial groups studied.

INDICATIONS

EFAVIR (efavirenz) Tablets, in combination with other antiretroviral agents, are indicated for HIV-1 infection.

DOSAGE AND METHOD OF ADMINISTRATION

Adult

Combined with other antiretrovirals: 600 mg once daily. Dosing at bedtime recommended during 1st 2-4 wk of therapy to improve tolerability.

Child

Combined with other antiretrovirals: >3 yr, 10-14 kg: 200 mg; 15-19 kg: 250 mg; 20-24 kg: 300 mg; 25-32.4 kg: 350 mg; 32.5-39 kg: 400 mg; ≥40 kg: 600 mg. To be taken once daily.

CONTRAINDICATIONS

Hypersensitivity

EFAVIR Tablets are contraindicated in patients with previously demonstrated clinically significant hypersensitivity (eg, Stevens-Johnson syndrome, erythema multiforme, or toxic skin eruptions) to any of the components of this product.

Contraindicated Drugs

For some drugs, competition for CYP3A by efavirenz could result in inhibition of their metabolism and create the potential for serious and/or life-threatening adverse reactions (eg, cardiac arrhythmias, prolonged sedation, or respiratory depression). Drugs that are contraindicated with efavirenz are listed in Table 2.

Table 2: Drugs That Are Contraindicated or Not Recommended for Use with Efavirenz

Drug Class: Drug Name	Clinical Comment
Antimigraine: ergot derivatives (dihydroergotamine, ergonovine, ergotamine, methylergonovine)	Potential for serious and/or life-threatening reactions such as acute ergot toxicity characterized by peripheral vasospasm and ischemia of the extremities and other tissues.
Benzodiazepines: midazolam, triazolam	Potential for serious and/or life-threatening reactions such as prolonged or increased sedation or respiratory depressions.
Calcium channel blocker: Bepridil	Potential for serious and/or life-threatening reactions such as cardiac arrhythmias.
GI motility agent: cisapride	Potential for serious and/or life-threatening reactions such as cardiac arrhythmias.
Neuroleptic: pimozide	Potential for serious and/or life-threatening reactions such as cardiac arrhythmias.
St. Joh's wort (<i>Hypericum perforatum</i>)	May lead to loss of virologic response and possible resistance to efavirenz or to the class of non-nucleoside reverse transcriptase inhibitors (NNRTI)

WARNINGS AND PRECAUTIONS

Mild to moderate liver disease; renal impairment. Monitor liver enzymes and cholesterol. Discontinue if severe skin rash or fever develops; known or suspected hepatitis B or infection. History of mental illness or seizures; elderly. Pregnancy. Children.

QTc prolongation has been observed with the use of efavirenz (see **Section PHARMACODYNAMICS and Section DRUG INTERACTIONS**). Consider alternatives to Efavir when coadministered with a drug with a known risk of *Torsade de Pointes* or when administered to patients at higher risk of *Torsade de Pointes*.

Nervous System Symptoms

Late-onset neurotoxicity, including ataxia and encephalopathy (impaired consciousness, confusion, psychomotor slowing, psychosis, delirium), may occur months to years after beginning efavirenz therapy. Some cases of late-onset neurotoxicity have occurred in patients with CYP2B6 genetic polymorphisms, which are associated with increased efavirenz levels despite standard dosing of Efavir. Patients presenting with signs and symptoms of serious neurologic adverse experiences should be evaluated promptly to assess the possibility that these events may be related to efavirenz use, and whether discontinuation of Efavir is warranted.

DRUG INTERACTIONS

Hormonal contraceptives; antibacterial e.g. rifampicin, rifabutin, clarithromycin; enzyme inducers. St. John's wort may decrease serum level. Ethanol (hepatic and CNS adverse effects).

Potentially Fatal: Life-threatening adverse effects when used with terfenadine, astemizole, cisapride, midazolam, triazolam and ergot alkaloids.

QT Prolongation Drugs

There is limited information available on the potential for a pharmacodynamic interaction between Efavir and drugs that prolong the QTc interval. QTc prolongation has been observed with the use of efavirenz (see Section PHARMACODYNAMICS and Section WARNINGS AND PRECAUTIONS). Consider alternatives to Efavir when coadministered with a drug with known risk of *Torsade de Pointes*.

Hepatic Impairment

The pharmacokinetics of efavirenz has not been adequately studied in patients with hepatic impairment. Because of the extensive CYP450-mediated metabolism of efavirenz and limited

clinical experience in patients with hepatic impairment, caution should be exercised in administering efavirenz to these patients (see **WARNINGS AND PRECAUTIONS**).

Renal Impairment

The pharmacokinetics of efavirenz has not been studied in patients with renal impairment. However, less than 1% of efavirenz is excreted unchanged in the urine, so the impact of renal impairment on efavirenz elimination should be minimal.

PREGNANCY

Pregnancy Category D

Efavirenz may cause fetal harm when administered during the first trimester to a pregnant woman. Pregnancy should be avoided in women receiving efavirenz. Barrier contraception should always be used in combination with other methods of contraception (eg, oral or other hormonal contraceptives). Because of the long half-life of efavirenz, use of adequate contraceptive measures for 12 weeks after discontinuation of efavirenz is recommended. Women of childbearing potential should undergo pregnancy testing before initiation of efavirenz. If this drug is used during the first trimester of pregnancy, or if the patient becomes pregnant while taking this drug, the patient should be apprised of the potential harm to the fetus. There are no adequate and well-controlled studies in pregnant women. Efavirenz should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus, such as in pregnant women without other therapeutic options. Malformations have been observed in 3 of 20 fetuses/infants from efavirenz-treated cynomolgus monkeys (versus 0 of 20 concomitant controls) in a developmental toxicity study. The pregnant monkeys were dosed throughout pregnancy (post-coital days: 20–150) with efavirenz 60 mg/kg daily, a dose that resulted in plasma drug concentrations similar to those in humans given 600 mg/day of efavirenz. Anencephaly and unilateral anophthalmia were observed in one fetus, microphthalmia was observed in another fetus, and cleft palate was observed in a third fetus. Efavirenz crosses the placenta in cynomolgus monkeys and produces fetal blood concentrations similar to maternal blood concentrations. Efavirenz has been shown to cross the placenta in rats and rabbits and produces fetal blood concentrations of efavirenz similar to maternal concentrations. An increase in fetal resorptions was observed in rats at efavirenz doses that produced peak plasma concentrations and AUC values in female rats equivalent to or lower than those achieved in humans given 600 mg once daily of efavirenz. Efavirenz produced no reproductive toxicities when given to pregnant rabbits at doses that produced peak plasma concentrations similar to and AUC values approximately half of those achieved in humans given 600 mg once daily of efavirenz.

Lactation

The Centers for Disease Control and Prevention recommend that HIV infected mothers should not breastfeed their infants to avoid risking postnatal transmission of HIV. Although it is not known if efavirenz is secreted in human milk, efavirenz is secreted into the milk of lactating rats. It has been recommended that HIV-infected women do not breastfeed their infants in order to avoid transmission of HIV. Because of the potential for HIV transmission and the potential for serious adverse effects in nursing infants, mothers should be instructed not to breastfeed if they are receiving efavirenz.

UNDESIRABLE EFFECTS

Rash including Stevens-Johnson syndrome and erythema multiforme; CNS effects e.g. dizziness, headache, insomnia, somnolence, abnormal dreams, fatigue, impaired concentration. Nausea, less frequently, vomiting, diarrhoea, hepatitis, depression, anxiety, psychosis, amnesia and ataxia, stupor, vertigo, abdominal pain, hepatic failure, pancreatitis, convulsions, gynaecomastia, pruritus, blurred vision.

Postmarketing Experiences: encephalopathy

OVERDOSE

Some patients accidentally taking 600 mg twice daily have reported increased nervous system symptoms. One patient experienced involuntary muscle contractions.

Treatment of overdose with efavirenz should consist of general supportive measures, including monitoring of vital signs and observation of the patient's clinical status. Administration of activated charcoal may be used to aid removal of unabsorbed efavirenz. There is no specific antidote for overdose with efavirenz. Since efavirenz is highly protein-bound, dialysis is unlikely to significantly remove the drug from blood.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

EFAVIR may cause dizziness, impaired concentration, and / or drowsiness. Patients should be instructed that if they experience these symptoms they should avoid potentially hazardous tasks such as driving or operating machinery.

STORAGE AND HANDLING INSTRUCTIONS

Store below 30°C away from moisture.

SHELF-LIFE

36 months

PACK SIZE

Carton containing HDPE bottle of 30 tablets each.

REGISTRATION HOLDER IN MALYASIA

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